Separating Neural Scale Response and Score Memory in Online Interval Calibration

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Working draft for student review, 11 October 2026

Abstract

Normalized-score calibration combines a neural forecaster's current scale with historical forecast errors divided by their originally issued scales. We study these two pathways separately in paired volatility-pulse simulations, keeping trained weights fixed and using the same observations across configurations. An initial five-seed GRU/TCN comparison is sensitive to training randomness. We replicate with three new training datasets and five paired seeds per dataset, retaining both current-scale and archived-score decomposition orders. The entry architecture contrast keeps its direction across new datasets, but training-randomness sensitivity still includes zero; the recovery contrast does not replicate reliably. In the reference Gaussian entry comparison, TCN has greater current-channel seed variability than GRU, while archive variability can dominate within a dataset. These phase-specific findings do not establish a general architecture ranking or coverage guarantee. Validation probes and independent audits accompany the complete results.

1 Research question and contribution scope

Financial uncertainty intervals should expand when risk rises and contract when a temporary disturbance ends. A learned scale model and an online calibrator can each retain information about that disturbance. Their combination may behave differently from either component alone. We ask: *where does variability across fitted neural models enter normalized-score calibration at shock entry and recovery?*

This controlled deep-learning study learns scales from raw past returns; a zero center isolates uncertainty adaptation. It examines a computational pathway without establishing trading gains or deployable conditional coverage.

Underspecification already explains why strong in-domain validation need not constrain deployment behavior [1]; HARNet studies initialization stability in neural volatility prediction [2]. Neural conformal control [3], jointly adapting scores and thresholds [4], retrospective residual updates [5], strongly adaptive experts [6], and scorecasting with feedback [7] are established. We do not claim these ideas as new. AdaptNC updates score parameters with replay; RetroAdj recomputes leave-one-out residuals. Our experiment fixes neural weights and separately varies current and archive contexts.

The intended increment is an explicit separation of *current neural scale* and *historically issued score denominators*, with paired observations, two decomposition orders, and independent training-data repeats. The initial crossed analysis was exploratory; its specific replication questions were fixed before new training and test generation. Evidence that fails to replicate must narrow the conclusion.

2 Crossed pathways and evaluation

For a fixed trained model, let $s_t^a > 0$ be the scale predicted from the last $a \in \{16, 64\}$ observations, excluding y_t. Define

$$q_t^{b,W} = \mathrm{OrderStat}_k \left\{ \frac{|y_i|}{s_i^b} : i = t - W, \ldots, t - 1 \right\}, \quad h_t^{a,b,W} = s_t^a q_t^{b,W}, \quad k = \lceil (W+1)(1-\alpha) \rceil. \tag{1}$$

Here $b \in \{16, 64\}$, $W \in \{32, 128\}$ and $\alpha = .1$. The deployed diagonal has $a = b$; the full eight-cell factorial changes the current and archive contexts independently. Warmup uses the last 128 scores of a separate validation path, with predictor-specific issued denominators.

For each cell and phase, $E(a, b, W)$ is its event-minus-matched-control mean interval score. Write $D(a, b) = E(a, b, 128) - E(a, b, 32)$. The diagonal context-by-window interaction is $J = D(64, 64) - D(16, 16)$, and

$$J = \underbrace{D(64, 64) - D(16, 64)}_{C_{b64}} + \underbrace{D(16, 64) - D(16, 16)}_{A_{a16}}, \tag{2}$$

$$= \underbrace{D(64, 16) - D(16, 16)}_{C_{b16}} + \underbrace{D(64, 64) - D(64, 16)}_{A_{a64}}. \tag{3}$$

These are exact computational differences, not unique causal mediation fractions. Correlated component interval endpoints cannot be added. C and A are channel-by-window interaction components, not channel main effects.

Training and tests. The GRU has 2041 parameters; the causal dilated TCN has 1897 and receptive field 64. Both use features $(y/3, \log(1+|y|))$, Gaussian scale loss, two contexts, data and paired sampler prefixes. Each target resets state. Adam and validation early stopping are fixed; total optimization steps may differ between architectures. Context length is not effective learned memory. Training uses 256 paths of length 1024; validation uses 64 paths and 8192 shared sampled targets. Each of three new datasets trains five seed pairs. The original five seed indices are reused: independence refers to new training/validation and test data, not 15 independent initialization/sampling draws.

The initial frozen grid has 30 cases: Gaussian, unit-variance t5 and stochastic volatility, five pulse durations and two amplitudes. The new replication fixes Gaussian/t5, amplitude 3 and durations 32/128, with 128 independent paired paths per family. Pulses begin at 384; recovery64 follows their exit. Families have separate noise streams; durations share their family stream, so both duration conditions have identical entry32 and are not independent replications. Event labels and true scales enter generation and evaluation only, never the model or quantile.

Scores and uncertainty. We report interval score $2h + (2/\alpha)(|y| - h)_+$ [8], coverage, width, oracle coverage deficit and excess width. The two reference contrasts are GRU J minus TCN J for Gaussian D128/A3, at entry and recovery64. The 4000-draw paired-path bootstrap averages all 15 fixed fitted pairs before resampling; 97.5% marginal intervals provide a two-phase Bonferroni approximation. Crossed dataset/seed/path resampling is sensitivity only: shared seed draws stay shared across datasets, with only three datasets and five seeds. Secondary intervals are descriptive. Finite quantile ranks also differ: ideal continuous exchangeable-score coverage is 30/33 for W32 and 117/129 for W128. The contrast therefore includes rank discretization as well as history length; this benchmark is not a coverage guarantee for our dependent scores.

3 Results and interpretation

Initial study motivated replication. In the initial five-seed Gaussian D128/A3 study, GRU-minus-TCN J is -0.05287 at entry and $+0.03276$ at recovery64. Fixed-fit 97.5% intervals exclude zero, but crossed seed/path intervals include zero. The entry seed range is $[-0.42866, +0.47620]$; omitting one seed can reverse the mean.

Independent training-data replication. With 15 new paired fits, the reference contrasts are -0.1986 at entry, fixed-fit 97.5% interval $[-0.2494, -0.1485]$, and $+0.0069$ at recovery64, interval $[-0.0052, +0.0192]$. Crossed sensitivity ranges are $[-0.4255, +0.0179]$ and $[-0.0589, +0.0673]$. All initialization outcomes remain in the analysis; these are interactions, not absolute interval-score rankings. Entry has the same negative mean direction in all three datasets, but the crossed interval includes zero. Recovery changes direction across datasets; omitting panel 3 reverses its mean. The earlier positive recovery contrast therefore did not replicate reliably.

Phase	Panel 1	Panel 2	Panel 3
Entry	-0.3101	-0.2547	-0.0309
Recovery64	-0.0110	-0.0107	+0.0425

Entries average five paired seeds per new training-data panel. Shared seeds and test paths preclude six independent significance tests.

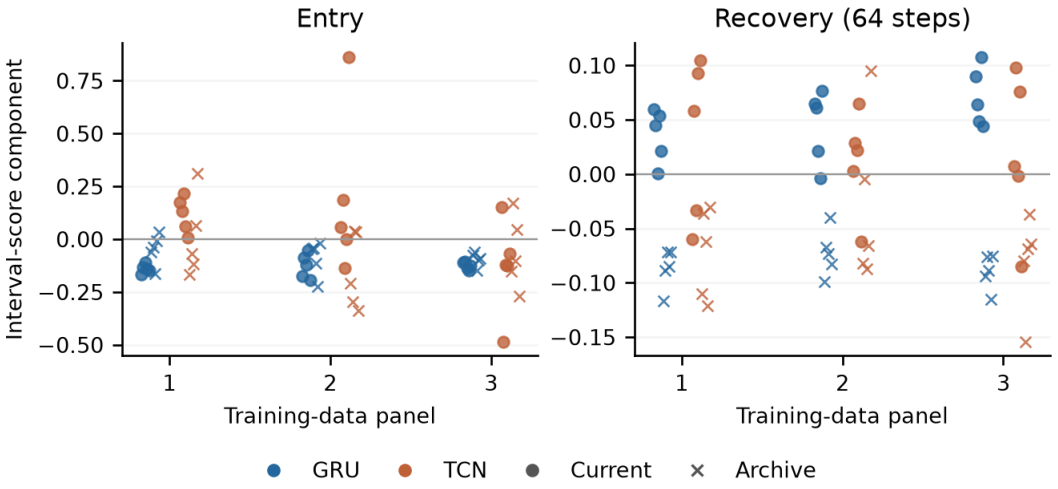


Figure 1: All five seeds per architecture and new dataset, Gaussian D128/A3. Current circles denote C_{b64} and archive crosses A_{a16}. Point spread describes training variability, not causal variance attribution.

Model	Phase	Total	Current	Archive
GRU	Entry	-0.2071	-0.1302	-0.0769
GRU	Recovery64	-0.0326	+0.0504	-0.0830
TCN	Entry	-0.0085	+0.0611	-0.0697
TCN	Recovery64	-0.0396	+0.0210	-0.0606

The table uses $C_{b64} + A_{a16}$. TCN entry current seed standard deviation exceeds GRU's in all six panel-by-route comparisons, but exceeds its own archive spread in only $4/6$. Panel 1 has larger archive spread in both routes, limiting claims of current dominance. These comparisons are correlated and descriptive; all routes, conditions, individual effects and absolute metrics are saved.

4 Diagnostics, limitations, and remaining work

Implementation checks are separate from discoveries. The new evaluation verifies 480 diagonal cells and 960 crossed finite-memory controls. If event/control observations agree from exit e, a fixed current context a, archive context b and window W give identical half-widths from $e + \max(a, b + W)$. This follows directly from input dependency; it is an implementation check, not a new theorem or a claim that real markets recover within 192 days. Persistent online controller states need not obey this bound. Full-sequence TCN64 execution is checked against the original reset-window predictor on validation before test access; canonical pulse horizons preserve numerically identical entry prefixes.

Independent audits reproduce all 30 selected validation losses, sampling digests, stopping decisions and response features. These checks establish implementation consistency, not scientific novelty.

A score interaction is not a memory-quality label. For Gaussian $Y \sim N(0, \tau^2)$, expected interval score is

$$R(h; \tau) = 2h + \frac{4}{\alpha}\{\tau\phi(h/\tau) - h\,\overline{\Phi}(h/\tau)\}.$$

Its mixed derivative for $h = sq$ is $R_h + hR_{hh} = 2 - (4/\alpha)\{\overline{\Phi}(z) - z\phi(z)\}$, which changes sign. A fixed-archive scale-by-quantile contrast is its oriented rectangle integral. Consequently, signed J combines response, archive quantiles and scoring nonlinearity; it cannot by itself diagnose harmful memory. These are standard calculus explanations, not a novel scoring rule. In the already-seen seed 4101 Gaussian/t5 reference, integrating current innovations preserves the main interaction directions, but does not resolve initialization variability.

Absolute costs accompany interaction effects. The saved Gaussian D128/A3 recovery64 means below use context 64 and all 15 fits per architecture. W128 has higher coverage and width than W32, but also higher score; these descriptive means are not another primary test.

Model	Interval	Score	Coverage	Width
GRU	Native	4.986	94.42%	4.544
GRU	W32	5.245	91.48%	4.439
GRU	W128	5.299	95.09%	4.902
TCN	Native	4.938	94.26%	4.470
TCN	W32	5.229	91.66%	4.452
TCN	W128	5.283	95.16%	4.887

Validation-response probes are exploratory. Each new training panel uses its first 256 saved validation targets for raw-input gradients and older-block permutation, plus 64 separate Gaussian validation pulses for log-scale response. All 30 fitted models are retained in NLL/gradient/pulse-response scatterplots. Validation distributions differ across panels, so plots separate panels. No equivalence test, regression, predictive meta-model or model selection has been performed. Gradients are local, permutation may create out-of-distribution contexts, and controlled pulses are stress probes, not online knowledge of a market event.

Financial application and boundaries. An earlier public ECB pilot uses USD/JPY/GBP/CHF reference rates per EUR: training through 2018, validation 2019–2020 and exploratory 2021–2026 evaluation. It does not clearly establish GRU over EWMA, and the four currencies share a denominator and dates. This already-viewed period is not an untouched financial holdout. The main result remains a synthetic, zero-center scale study with small neural models, one training generator and one optimization procedure. No private competition data are used.

The present probes do not yet justify a predictive response model: within-architecture patterns vary across datasets. A future prediction test would need a common diagnostic dataset, architecture-plus-NLL controls, and entirely new training datasets held out. We first consolidate the mechanism evidence and its failures. New adaptation methods would require validation selection, direct prior-method comparisons and another fresh test. This is a working draft for student review.

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Reproducibility and draft status

Frozen protocols, complete metrics, checkpoints, source/input hashes, independent audits and all unsuccessful conditions are available at https://github.com/Lebmar/DL-project/tree/codex/stage2-memory-study. Detailed runbooks distinguish existing-output verification from a new run; writers refuse silent overwriting. The report's numbers and compact figure are rendered from the saved Stage3 analysis without re-fitting or re-bootstrap. Main text ends on page 4; references and this note are extra. AI tools assisted implementation, checking and draft preparation with course permission. The student must understand, revise and take responsibility for the work and replace the author line with the required official name before submission.